

Theorem 2.11. *Every 4-potent ring is commutative.*

Proof. Let R be 4-potent. By Lemma 2.5, $2x = 0$ for all $x \in R$. Thus $(x^2 + x)^2 = x^4 + x^2 = x^2 + x$ using Lemma 2.10. By Lemma 2.2, $x^2 + x$ is central for all $x \in R$, and so R is commutative by Lemma 2.9. $\square$

Theorem 2.12. *Every 5-potent ring is commutative.*

Proof. Let R be 5-potent. By Lemma 2.4, $x^2 \in Z(R)$ for all $x \in R$. Thus $(x^2 + x)^2 - x^4 - x^2 = 2x^3$ is central for all $x \in R$. Next $2x = 2x^5 = 2x^3 \cdot x^2 \in Z(R)$ for all x . Finally,

$$\begin{aligned} x^2 + x &= (x^2 + x)^5 \\ &= x^{10} + 5x^9 + 10x^8 + 10x^7 + 5x^6 + x^5 \\ &= x^2 + 5x + 10x^4 + 10x^3 + 5x^2 + x \\ &= 2(5x^4 + 5x^3 + 3x^2 + 3x) \in Z(R). \end{aligned}$$

By Lemma 2.9, R is commutative. $\square$

We will postpone the case $n = 6$ for now.

Theorem 2.13. *Every 7-potent ring is commutative.*

Proof. Let R be 7-potent. By Lemma 2.4, $x^3 \in Z(R)$ for all $x \in R$. Thus $(x^2 + x)^3 - x^6 - x^3 = 3(x^5 + x^4) \in Z(R)$. Multiplying by x^3 , we get $3(x^2 + x) \in Z(R)$. It follows that

$$3((x + y)^2 + x + y) - 3(x^2 + x) - 3(y^2 + y) = 3(xy + yx) \in Z(R).$$

By Lemma 2.8 (with $k = 3$), $3x^2 \in Z(R)$. Thus $3x = 3(x^2 + x) - 3x^2 \in Z(R)$ for all $x \in R$.

Next, we have

$$\begin{aligned} x^2 + x &= (x^2 + x)^7 \\ &= x^{14} + 7x^{13} + 21x^{12} + 35x^{11} + 35x^{10} + 21x^9 + 7x^8 + x^7 \\ &= x^2 + 7x + 21x^6 + 35x^5 + 35x^4 + 21x^3 + 7x^2 + x. \end{aligned}$$

Cancelling $x^2 + x$ from both sides and separating terms, we get

$$0 = 3(2x + 7x^6 + 11x^5 + 11x^4 + 7x^3 + 2x^2) + x + 2x^5 + 2x^4 + x^2.$$

Thus $2x^5 + 2x^4 + x^2 + x \in Z(R)$. Replacing x in this last expression with $-x$ and adding gives $4x^4 + 2x^2 \in Z(R)$. Thus $x^4 - x^2 = 4x^4 + 2x^2 - 3(x^4 + x^2) \in Z(R)$.

Since both x^3 and $x^4 - x^2$ are central, so is $(x^4 - x^2)^2 + 2(x^3)^2 = x^8 + x^4 = x^2 + x^4$. Therefore $2x^2 = (x^2 + x^4) - (x^4 - x^2) \in Z(R)$, and so $x^2 = 3x^2 - 2x^2 \in Z(R)$.

Finally, $x = x^7 = x^2 \cdot x^2 \cdot x^3 \in Z(R)$ for all $x \in R$, and we are finished. $\square$

Many other cases of $x^n = x$ for small n can be handled by these methods, the more difficult cases being when n is odd (although Lemma 2.4 certainly helps) or a power of 2. We believe our proofs are close to being best possible (in some sense) and would be pleased to hear from any reader who can shorten or improve any of them.

We conclude this section with one more example, covering the cases $n = 6, 10, 18, 34, \dots$ in a single theorem.

Theorem 2.14. *Let $j \geq 1$ be an integer and assume that every 2^{j+1} -potent ring is commutative. Let $i > j$ be an integer and let R be a $2^i + 2^j$ -potent ring. Then R is commutative.*

Proof. We have $2x = 0$ for all $x \in R$ by Lemma 2.5. Now

$$\begin{aligned} x^2 + x &= (x^2 + x)^{2^i+2^j} &= (x^2 + x)^{2^i}(x^2 + x)^{2^j} \\ &= (x^{2^{i+1}} + x^{2^i})(x^{2^{j+1}} + x^{2^j}) &= x^{2(2^i+2^j)} + x^{2^i+2^{j+1}} + x^{2^{i+1}+2^j} + x^{2^i+2^j} \\ &= x^2 + x^{2^i+2^j+2^j} + x^{2^i+2^i+2^j} + x &= x^2 + x^{2^j+1} + x^{2^i+1} + x, \end{aligned}$$

using Lemma 2.10 in the third equality. Cancelling, we have $x^{2^j+1} + x^{2^i+1} = 0$, that is, $x^{2^i+1} = x^{2^j+1}$. Finally, we have

$$x = x^{2^i+2^j} = x^{2^i+1}x^{2^j-1} = x^{2^j+1}x^{2^j-1} = x^{2^j+1}$$

for all $x \in R$. By assumption, R is commutative. $\square$

Corollary 2.15. *Let R be an (2^i+2) -potent ring where $i > 1$. Then R is commutative.*

Proof. Take $j = 1$ in Theorem 2.14 and apply Theorem 2.11. $\square$

3. HERSTEIN'S THEOREM FOR $n = 2, 4, 8$

In a ring R , we have already used ring commutators $[x, y] = xy - yx$ and will do so quite heavily in this section. Among the properties satisfied by commutators, we will need the following:

$$[x, x] = 0, \quad [x, y] = -[y, x], \quad [x, y + z] = [x, y] + [x, z]$$

for all $x, y, z \in R$. To improve readability, it is useful to introduce, for each $x \in R$, the mapping $\text{ad}(x) : R \rightarrow R$ defined by $\text{ad}(x)(y) = [x, y]$ for all $y \in R$. Properties satisfied by this mapping include:

$$\begin{aligned} \text{ad}(x)(y + z) &= \text{ad}(x)(y) + \text{ad}(x)(z), \\ \text{ad}(x + y) &= \text{ad}(x) + \text{ad}(y), \\ \text{ad}([x, y]) &= \text{ad}(x)\text{ad}(y) - \text{ad}(y)\text{ad}(x). \end{aligned}$$

for all $x, y, z \in R$. The first identity says that $\text{ad}(x)$ is an endomorphism of the underlying abelian group $(R, +)$. The second says that $\text{ad} : (R, +) \rightarrow \text{End}(R, +)$ is a homomorphism of abelian groups. The third is where the symbol “ad” comes from; the identity says that $\text{ad} : (R, [\cdot, \cdot]) \rightarrow \text{End}(R, +)$ is the *adjoint* representation of the Lie ring associated to R . However, for us, ad is just a notational shorthand; nothing about a ring's Lie ring plays a role outside of just using the identities above in calculations.

As discussed in §1, we are interested in rings R satisfying $[x^n - x, y] = 0$ for some integer $n > 1$ and all $x, y \in R$. We are specifically interested in $n = 2, 4$ or 8 , so we will start with the broader assumption that

$$[p(x) - x, y] = 0 \quad \text{for all } x, y \in R \tag{E}$$

where $p(t)$ is a polynomial with integer coefficients such that $p(t) = q(t^2)t^2$, where $q(t) \in \mathbb{Z}[t]$.

Lemma 3.1. *Let R be a ring satisfying (E). Then $2\text{ad}(x) = 0$ for all $x \in R$.*

Proof. For all $x \in R$, $-\text{ad}(x) = \text{ad}(-x) = \text{ad}(p(-x)) = \text{ad}(p(x)) = \text{ad}(x)$. $\square$

Roughly speaking, the conclusion of Lemma 3.1 is, in the present setting, what replaces R having characteristic 2 in the n -potent setting for even n . The conclusion can be stated in various equivalent ways, such as $[x, y] = [y, x]$ for all $x, y \in R$.

Lemma 3.2. *Let R be a ring such that $2\text{ad}(x) = 0$ for all $x \in R$. Then for each nonnegative integer n ,*

$$\text{ad}(x)^{2^n} = \text{ad}(x^{2^n}).$$